Comparative Safety of Long-Acting Versus Short-Acting Erythropoiesis-Stimulating Agents (ESA): Disproportionality Analysis Using the FDA Adverse Event Reporting System

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Objectives: Erythropoiesis-stimulating agents (ESAs), classified as long-acting or short-acting, is a primary treatment for anemia in chronic kidney disease and chemotherapy. Given ongoing safety concerns regarding cardiovascular events and potential immunogenic risks associated with PEGylation, this study aims to detect safety signals for long-acting versus short-acting ESAs.

Methods: We conducted disproportionality analyses using the FDA Adverse Event Reporting System (FAERS) database. All adverse event (AE) reports related to long-acting and short-acting ESA use from Q1 2018 to Q4 2022 were analyzed, including darbepoetin alfa and methoxy polyethylene glycol-epoetin beta as long-acting ESAs and epoetin alfa and epoetin beta as short-acting ESAs. Data were divided into 2 data sets: ESA as the primary suspect (PS) and ESA as either the PS or secondary suspect (SS). Proportional reporting ratios, reporting odds ratios, and 95% CIs for information components were calculated to identify safety signals. Sensitivity analyses included AE reports with unspecified ESA types to confirm signal consistency.

Results: A total of 5702 and 9772 reports for long-acting ESAs, 2487 and 3975 for short-acting ESAs, and 2 and 352 for ESAs with unspecified drug names that could not be classified as either long-acting or short-acting were identified in the PS and PS/SS data sets. In the PS-restricted primary analysis, 10 robust safety signals were identified for long-acting ESAs: back pain, chest pain, drug hypersensitivity, dyspnea, falls, hypotension, pruritus, urinary tract infection, urticaria, and vomiting. Notably, signals such as falls and hypotension were uniquely detected when isolating the PS role and failed to reach the signal threshold in the broader secondary analysis.

Conclusions: Our analysis found disproportionately reported AEs for long-acting ESAs, particularly when ESAs were the PS, highlighting the importance of monitoring AEs not listed in the package label, such as falls. Furthermore, despite existing black box warnings regarding cardiovascular risks, our analysis found weak evidence for cardiovascular safety signals strictly attributable to long-acting ESAs. Further studies are needed to confirm the association between these signals and long-acting ESAs.

Key Words: erythropoiesis-stimulating agents, adverse drug events, safety signal, disproportionality analysis

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Erythropoiesis-stimulating agents (ESAs) are recombinant versions of erythropoietin (EPO), a glycoprotein hormone produced by the peritubular cells of the kidney. ESAs that acts like endogenous EPO stimulate the bone marrow to produce red blood cells.[1] Because EPO production decreases as kidney function declines, ESAs are used to treat anemia caused by chronic kidney disease (CKD) and manage iron deficiency anemia in CKD patients on dialysis.[2,3] In addition, ESAs are indicated for chemotherapy-associated anemia among patients with cancer.[4,5]

ESAs are classified into long-acting and short-acting ESAs based on their duration of action. Epoetin alfa and beta are considered as short-acting ESAs, with relatively short half-lives of 6 to 8 hours. In contrast, long-acting ESAs, such as darbepoetin alfa and methoxy polyethylene glycol-epoetin beta—also known as a continuous erythropoiesis receptor activator (CERA)—feature significantly extended half-lives of up to 130 hours. While the prolonged half-life allows for less frequent dosing and more stable hemoglobin levels, it may also introduce distinct safety concerns.[6–9]

Crucially, the sustained and intense stimulation of erythropoietin receptors (EPOR) on nonerythroid cells presents a potential pathway for patient harm. EPOR is widely expressed in diverse tissues, including endothelial cells, platelets, and vascular smooth muscle cells, where its chronic activation may trigger cardiovascular (CV) complications such as hypertension and thrombosis.[10] Furthermore, recent evidence suggests that the PEGylation used in long-acting formulations may carry unique risks; despite the long-held belief that PEG is nonimmunogenic, it can induce anti-PEG antibodies, leading to hypersensitivity reactions or "silent inactivation" of the drug. These pharmacokinetic and pharmacodynamic differences suggest that long-acting and short-acting ESAs may have divergent safety profiles, yet comparative clinical evidence remains insufficient.[11]

The unique pharmacokinetic and pharmacodynamic characteristics of ESA formulations may result in different safety profiles between long-acting and short-acting ESAs, but evidence on their relative risks remains poor.[12,13] The Cochrane group conducted a meta-analysis

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of randomized controlled clinical trials (RCT) to compare the safety of ESAs.[14] The safety outcomes of interest included CV death, myocardial infarction, stroke, hypertension, kidney failure, and vascular access thrombosis. The study concluded the current data from RCTs are insufficient to determine the comparative safety of different ESA formulations for the safety outcomes they focused on. This conclusion was attributed to the fact that most studies were at high risk of bias or lacked sufficient methodological clarity, which limits the reliability of the available evidence, as noted by the Cochrane group. Only a limited number of observational studies on the cardiac safety of ESAs are available, and data on other safety issues related to ESAs are scarce.[15]

Case reports compiled in the spontaneous adverse event (AE) reporting system can be used to comprehensively review the types and relatively higher reporting frequencies of certain AEs through data mining.[16] This is particularly important in CKD or cancer patients with anemia, where distinguishing between adverse effects of treatment and symptoms of the disease itself can be challenging. This study aims to identify overall safety profiles of long-acting ESAs compared with short-acting ESAs, utilizing the FDA's Adverse Event Reporting System (FAERS) database in the United States.

METHODS

Data Source

We analyzed spontaneous AE reports which had been collected in the FAERS database from January 2018 to December 2022. The FAERS database was established for the FDA's postmarketing drug safety surveillance. It is a publicly available database that collects AEs reported by health care providers and patients within and outside of the United States and used in the detection of safety signals through quantification of the association between drugs and AEs called disproportionality analysis.[17,18] As this study utilized publicly accessible data, it does not require ethical approval.

Data Integration and Table Definitions

For this study, we integrated 5 core tables from the FAERS database—DEMO, DRUG, INDI, OUTC, and REAC—linked by the unique report identifier (primaryid). We utilized quarterly data sets from the first quarter (Q1) of 2018 through the fourth quarter (Q4) of 2022 (downloaded November 28, 2023), with the final reports in the data set dated October 14, 2022.[19,20] The specific tables and the relevant columns utilized in our analysis are as follows:

DEMO (Demographic): Used for patient characteristics and deduplication. Relevant columns include *primaryid*, *caseid*, *fda_dt* (FDA receipt date), *age*, and *gndr_cod* (gender).

DRUG (Drug Information): Used to identify and categorize ESA exposures and their suspected roles. Relevant columns include *primaryid*, *prod_ai* (active ingredient), *drugname*, and *role_cod* (e.g., primary suspect).

INDI (Indication): Used to identify the clinical reason for ESA use. Relevant columns include *primaryid* and *indi_pt* (reported indication).

OUTC (Outcome): Used to characterize the severity of the adverse event. Relevant columns include *primaryid* and *outc_cod* (e.g., death, hospitalization).

REAC (Reaction): Used to identify specific AE types. Relevant columns include *primaryid* and *pt* (Preferred Terms coded according to MedDRA).[19]

Identification and Classification of AE Reports Related to ESAs

ESAs were classified into 3 categories—short-acting, long-acting, or uncertain—based on the active ingredient and brand name information extracted from the *prod_ai* and *drugname* fields in the DRUG table. The "uncertain" category was assigned to entries lacking specific formulation details, such as blinded clinical trial agents or generic terms (e.g., "EPO" and "Erythropoietin"). A detailed list of these classifications is provided in Supplementary Table 1, Supplemental Digital Content 1, http://links.lww.com/JPS/A821. Following classification, ESA reports were merged with the DEMO, INDI, and OUTC files using the *primaryid* as the matching variable. Before merging, the DEMO file was preprocessed to remove duplicate reports by excluding identical combinations of *caseid*, *primaryid*, and *fda_dt* as recommended by the FDA.[20]

Identification and Classification of AE Types From the AE Reports Related to ESAs

To identify specific AEs, the integrated data set was merged with the REAC table using the *primaryid*. AE types were extracted from the *pt* field, utilizing PT according to the MedDRA hierarchy.[19] To ensure data integrity, duplicate cases—defined as reports with identical *primaryid*, occurrence date, age, sex, country, and AE type—were excluded.

Analysis

We conducted a descriptive analysis of unique AE reports derived from a data set integrating DEMO, DRUG, INDI, and OUTC files. The analysis characterized the reports based on age, sex, drug role, reporting year, country of occurrence, reporter type, indications, and outcomes. Regarding drug role, each AE report could include multiple drugs identified as potentially associated with the AE, with each drug assigned 1 of 4 roles by the reporter: primary suspect (PS), secondary suspect (SS), concomitant (C), and interacting (I).

AE types, identified through further integration with the REAC file and represented as PTs, were classified into their respective system organ class (SOC) using the MedDRA hierarchy.[21] To assess differences in report characteristics and AE types among long-acting, short-acting, and uncertain type ESAs, Kruskal-Wallis and χ^2 tests were performed, with statistical significance set at P-value <0.05.

To evaluate safety signals for long-acting versus short-acting ESAs, we applied disproportionality analysis, a commonly used pharmacovigilance method to detect distribution imbalances between a drug of interest and the AE.[22–26] This approach included: (1) mapping drug-AE pairs into 2×2 contingency tables for long-acting versus short-acting ESAs, and (2) calculating disproportionality metrics: the reporting odds ratio (ROR), proportional reporting ratio (PRR) and the 95% 2-sided CI for information component (IC). The IC and its 95% CI were calculated by the Bayesian confidence propagation neural network (BCPNN) algorithm. Safety signals were defined using established thresholds for disproportionality analysis: the number of reports ≥ 3, PRR, and ROR ≥ 2 (with χ^2

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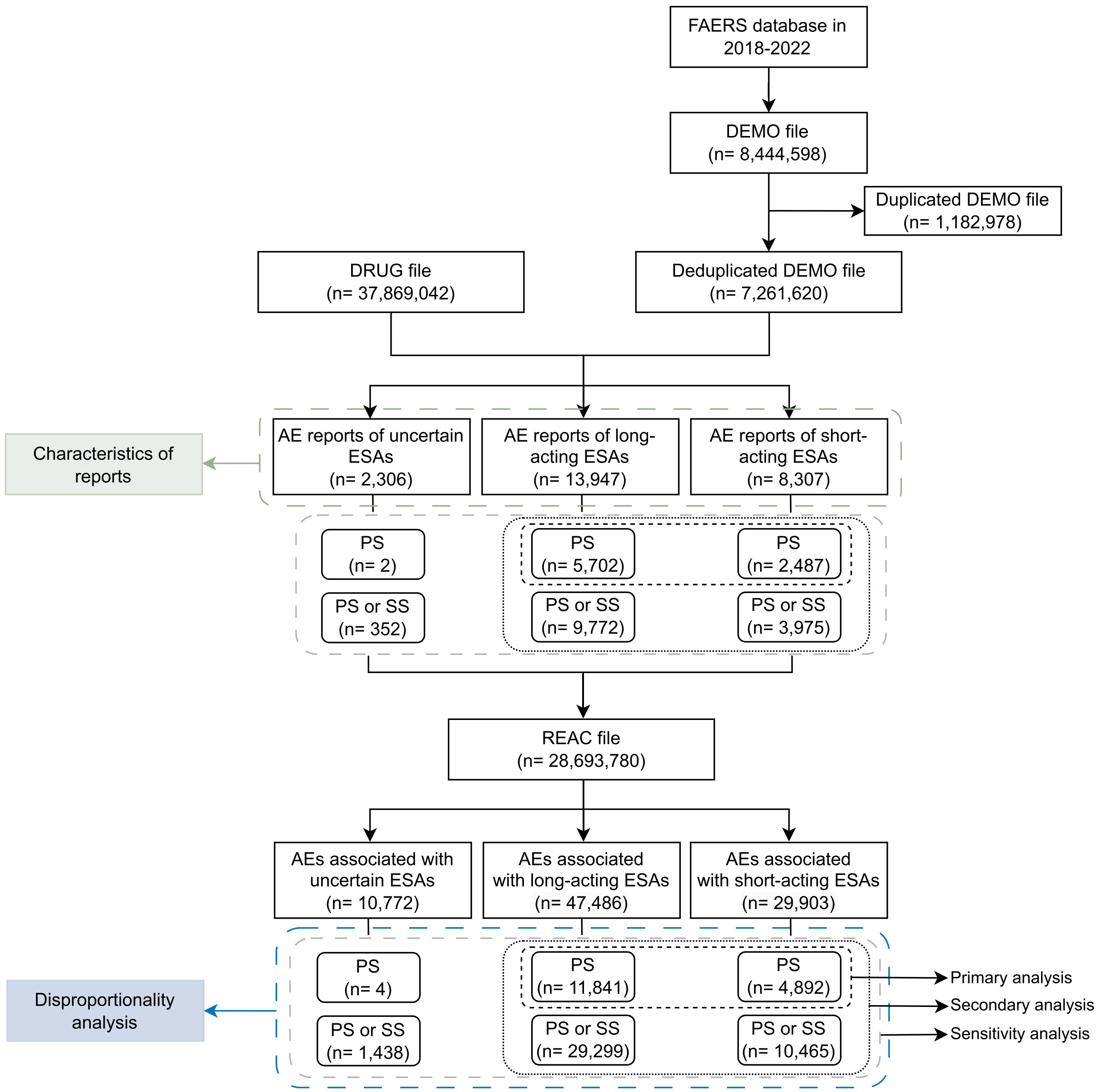

FIGURE 1. Flow diagram of selecting adverse event reports for disproportionality analysis. This figure outlines the step-by-step process of data preparation and classification for adverse events associated with erythropoiesis-stimulating agents using the FAERS database (2018-2022). DEMO indicates demographic files; DRUG, drug files; REAC, reaction files.

value ≥4 for both), and a lower limit of the 95% CIs for IC (IC025) >0.[27]

The primary analysis included AE reports where ESAs were assigned the role of PS, while the secondary analysis incorporated reports where ESAs were identified as either PS or SS. Sensitivity analyses were performed by reclassifying AE reports with uncertain ESA type as either long-acting or short-acting. This approach allowed to evaluate the consistency of safety signals in different classification scenarios.

Statistical analysis was performed using MySQL 8.0 and R ver. 4.3.3 (R foundation for Statistical Computing, Vienna, Austria). This study adhered to The REporting of A Disproportionality analysis for drUg Safety signal detection using individual case safety reports in PharmacoVigilance (READUS-PV) statement.[28,29]

RESULTS

Characteristics of Adverse Event Reports

Figure 1 illustrates the identification process of AE reports. Among the 8,444,598 reports in the FAERS database (2018-2022), we identified 13,947 reports associated with long-acting ESAs and 8307 with short-acting ESAs. As detailed in Table 1, patients using short-acting ESAs were slightly older (mean: 66.0, SD: 18.3) compared with those using long-acting ESAs (mean: 64.5, SD: 18.0). While reports originated predominantly from North

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TABLE 1. Description of AE Reports of ESAs Submitted to the FAERS From 2018 to 2022

Characteristics	Long-Acting ESAs (N = 13,947, 100%)	Short-Acting ESAs (N = 8307, 100%)	Uncertain Types of ESAs (N = 2306, 100%)	*P*
Age				
Mean ± SD	64.5 ± 18.0	66.0 ± 18.3	60.5 ± 18.7	< 0.001
Age group (%)				< 0.001
< 20	188 (1.3)	180 (2.2)	43 (1.9)	
20-29	304 (2.2)	118 (1.4)	115 (5.0)	
30-39	487 (3.5)	201 (2.4)	158 (6.9)	
40-49	829 (5.9)	278 (3.3)	132 (5.7)	
50-59	1389 (10.0)	693 (8.3)	261 (11.3)	
60-69	2083 (14.9)	1215 (14.6)	420 (18.2)	
70-79	2351 (16.9)	1484 (17.9)	467 (20.3)	
≥ 80	2076 (14.9)	1191 (14.3)	238 (10.3)	
Unknown	4240 (30.4)	2947 (35.5)	472 (20.5)	
Sex (%)				< 0.001
Male	6479 (46.5)	3311 (39.9)	1075 (46.6)	
Female	6045 (43.3)	3529 (42.5)	989 (42.9)	
Unknown	1425 (10.2)	1468 (17.7)	242 (10.5)	
Classification of drugs (%)				< 0.001
Primary suspect	5702 (40.9)	2487 (29.9)	2 (0.1)	
Primary or secondary suspect	9772 (70.1)	3975 (47.9)	352 (15.3)	
Primary, secondary suspect, or concomitant	13,946 (99.9)	8303 (99.9)	2301 (99.8)	
Primary, secondary suspect, concomitant, or interacting	13,947 (100.0)	8307 (100.0)	2306 (100.0)	
Reporting year (%)				< 0.001
2022	2796 (20.0)	1690 (20.3)	780 (33.8)	
2021	2210 (15.8)	1538 (18.5)	355 (15.4)	
2020	2334 (16.7)	1742 (21.0)	365 (15.8)	
2019	3088 (22.1)	1724 (20.8)	422 (18.3)	
2018	3495 (25.1)	1578 (19.0)	378 (16.4)	
Others	22 (0.2)	31 (0.4)	6 (0.3)	
Unknown	2 (0.0)	4 (0.0)	0	
Country of occurrence (%)				< 0.001
North America	5865 (42.1)	5669 (68.2)	355 (15.4)	
Europe	4734 (33.9)	1864 (22.4)	1011 (43.8)	
Asia	2367 (17.0)	327 (3.9)	437 (19.0)	
South America	103 (0.7)	39 (0.5)	387 (16.8)	
Others	688 (4.9)	68 (0.8)	48 (2.1)	
Unknown	190 (1.4)	340 (4.1)	68 (2.9)	
Reporter type (%)				< 0.001
Physician	3788 (26.8)	2133 (25.7)	881 (38.2)	
Other health professionals	3437 (24.6)	898 (10.8)	332 (14.4)	
Health professionals	3360 (24.1)	1508 (18.2)	588 (25.5)	
Consumer	2069 (14.8)	2043 (24.6)	360 (15.6)	
Pharmacist	1080 (7.7)	1166 (14.0)	108 (4.7)	
Lawyer	50 (0.4)	100 (1.2)	5 (0.2)	
Unknown	215 (1.5)	460 (5.5)	32 (1.4)	
Indication (%)*				< 0.001
Nephrogenic anemia	2005 (14.4)	514 (6.2)	95 (4.1)	
Anemia	1686 (12.1)	1269 (15.3)	392 (17.0)	
Chronic kidney disease	547 (3.9)	311 (3.7)	11 (0.5)	
Myelodysplastic syndrome	515 (3.7)	157 (1.9)	30 (1.3)	
End stage renal disease	77 (0.6)	65 (0.8)	0	
Hemoglobin decreased	64 (0.5)	90 (1.1)	19 (0.8)	
Prophylaxis	45 (0.3)	37 (0.4)	35 (1.5)	
Anemia prophylaxis	36 (0.3)	33 (0.4)	25 (1.1)	
Anemia of chronic disease	30 (0.2)	49 (0.6)	3 (0.1)	
Renal failure	30 (0.2)	25 (0.3)	1 (0.0)	
Outcome (%)				< 0.001
Other serious	5634 (40.4)	2685 (32.3)	856 (37.1)	
Hospitalization-initial or prolonged	4870 (34.9)	2592 (31.2)	861 (37.3)	
Death	2341 (16.8)	1159 (14.0)	395 (17.1)	
Life-threatening	540 (3.9)	323 (3.9)	100 (4.3)	
Disability	145 (1.0)	107 (1.3)	19 (0.8)	
Congenital anomaly	9 (0.1)	24 (0.3)	7 (0.3)	
Required intervention to prevent permanent impairment/damage	1 (0.0)	1 (0.0)	2 (0.1)	

*A total of 7006 (50.2%) for long-acting ESAs, 3474 (41.8%) for short-acting ESAs, and 1061 (46.0%) for uncertain type of ESAs were categorized as "Product used for unknown indication."

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TABLE 2. Frequency of Reported System Organ Class in Long-Acting ESAs and Short-Acting ESAs

System Organ Class*		Long-Acting ESAs	Short-Acting ESAs	P
(a) Primary suspect		(N = 11,841, 100%)	(N = 4892, 100%)	
General disorders and administration site conditions		2279 (19.2)	978 (20.0)	0.28
Injury, poisoning, and procedural complications		1440 (12.2)	1429 (29.2)	< 0.001
Surgical and medical procedures		876 (7.4)	263 (5.4)	< 0.001
Infections and infestations		840 (7.1)	190 (3.9)	< 0.001
Investigations		689 (5.8)	418 (8.5)	< 0.001
Gastrointestinal disorders		652 (5.5)	176 (3.6)	< 0.001
Immune system disorders		631 (5.3)	27 (0.6)	< 0.001
Nervous system disorders		624 (5.3)	214 (4.4)	0.017
Respiratory, thoracic, and mediastinal disorders		614 (5.2)	110 (2.2)	< 0.001
Skin and subcutaneous tissue disorders		583 (4.9)	112 (2.3)	< 0.001
Vascular disorders		472 (4.0)	119 (2.4)	< 0.001
Cardiac disorders		434 (3.7)	159 (3.3)	0.20
Blood and lymphatic system disorders		308 (2.6)	134 (2.7)	0.65
Renal and urinary disorders		277 (2.3)	84 (1.7)	0.014
Musculoskeletal and connective tissue disorders		251 (2.1)	126 (2.6)	0.08
Metabolism and nutrition disorders		223 (1.9)	74 (1.5)	0.11
Neoplasms benign, malignant, and unspecified (incl cysts and polyps)		190 (1.6)	63 (1.3)	0.15
(b) Primary or secondary suspect	Change in ranking	(N = 29,299, 100%)	(N = 10,465, 100%)	
General disorders and administration site conditions	—	4356 (14.9)	1826 (17.4)	< 0.001
Gastrointestinal disorders	↑4	2926 (10.0)	569 (5.4)	< 0.001
Injury, poisoning, and procedural complications	↓1	2833 (9.7)	2239 (21.4)	< 0.001
Investigations	↑1	2291 (7.8)	914 (8.7)	0.003
Infections and infestations	↓1	1902 (6.5)	478 (4.6)	< 0.001
Immune system disorders	↑1	1897 (6.5)	91 (0.9)	< 0.001
Nervous system disorders	↑1	1463 (5.0)	603 (5.8)	0.003
Cardiac disorders	↑4	1445 (4.9)	332 (3.2)	< .001
Respiratory, thoracic, and mediastinal disorders	—	1381 (4.7)	282 (2.7)	< 0.001
Skin and subcutaneous tissue disorders	—	1375 (4.7)	350 (3.3)	< 0.001
Surgical and medical procedures	↓8	1208 (4.1)	337 (3.2)	< 0.001
Vascular disorders	↓1	1190 (4.1)	307 (2.9)	< 0.001
Blood and lymphatic system disorders	—	887 (3.0)	502 (4.8)	< 0.001
Renal and urinary disorders	—	701 (2.4)	216 (2.1)	0.06
Metabolism and nutrition disorders	↑1	687 (2.3)	231 (2.2)	0.44
Musculoskeletal and connective tissue disorders	↓1	679 (2.3)	252 (2.4)	0.63
Neoplasms benign, malignant, and unspecified (incl cysts and polyps)	—	553 (1.9)	166 (1.6)	0.05

*Ordered by the frequency of AE types in reports of long-acting ESAs. Only system organ classes with a frequency > 1% are presented.
incl indicates including.

America and Europe, a notably higher proportion of short-acting ESA reports came from North America (68.2%) compared with long-acting ESAs (42.1%). In both groups, health care professionals, particularly physicians, were the most frequent reporters. The most frequent indications were related to anemia and CKD.

Frequent Adverse Events and System Organ Classes

Table 2 illustrates the distribution of AEs categorized by SOC for both the primary (ESA as the primary suspect, PS) and secondary (ESA as the primary or secondary suspect, PS/SS) analyses. In the primary analysis, the most frequently reported SOCs across both ESA formulations were predominantly associated with administrative or procedural complications, specifically general disorders and administration site conditions, injury, poisoning and procedural complications, and surgical and medical procedures. Beyond these procedural events, the reported AEs spanned a diverse array of organ systems. However, isolating the PS role revealed distinct clinical patterns; notably, long-acting ESAs exhibited higher reporting proportions in gastrointestinal disorders (5.5% versus 3.6%), immune system disorders (5.3% versus 0.6%), and respiratory, thoracic, and mediastinal disorders (5.2% versus 2.2%) compared with their short-acting counterparts.

Transitioning to the PT level, Table 3 details specific AEs reported with a frequency exceeding 1%. Severe clinical outcomes, specifically death and hospitalization, were highly reported across both the long-acting and short-acting ESA cohorts. Both groups also frequently shared procedural or medication-related issues, such as product storage errors and off-label use. However, when examining specific clinical symptoms, the profiles diverged notably, closely mirroring the SOC-level distinctions. The long-acting ESA cohort uniquely exhibited higher reporting frequencies for immune-related events (drug hypersensitivity and pruritus), respiratory symptoms (dyspnea and pneumonia), and gastrointestinal issues (nausea and vomiting). Furthermore, falls and urinary tract infections (UTI) emerged as distinct and prominent AEs predominantly associated with long-acting ESAs.

Safety Signals From Disproportionality Analysis

The results of the disproportionality analyses evaluating 1424 AE types for long-acting ESAs are detailed in Table 4. When strictly isolating ESAs as the PS, we

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TABLE 3. The Most Frequently Reported Adverse Events Types in Long-Acting and Short-Acting ESAs

Long-Acting ESAs		Short-Acting ESAs	
(a) Primary suspect			
Preferred term (N = 11,841, 100%)		Preferred term (N = 4892, 100%)	
Death	1013 (8.6)	Product storage error	470 (9.6)
Drug hypersensitivity	551 (4.7)	Death	406 (8.3)
Hospitalization	512 (4.3)	Off label use	210 (4.3)
Product storage error	324 (2.7)	Circumstance or information capable of leading to medication error	196 (4.0)
Dyspnea	323 (2.7)	Hemoglobin decreased	108 (2.2)
Pruritus	223 (1.9)	Hospitalization	107 (2.2)
Circumstance or information capable of leading to medication error	181 (1.5)	Drug ineffective	56 (1.1)
Nausea	178 (1.5)	Product dose omission	54 (1.1)
Urinary tract infection	162 (1.4)	Intercepted product administration error	52 (1.1)
Fall	151 (1.3)	Anemia	51 (1.0)
Off label use	147 (1.2)	Wrong technique in product usage process	49 (1.0)
Vomiting	132 (1.1)		
Hemoglobin decreased	121 (1.0)		
Pneumonia	114 (1.0)		
(b) Primary or secondary suspect			
Preferred term (N = 29,299, 100%)		Preferred term (N = 10,465, 100%)	
Drug hypersensitivity	1740 (5.9)	Product storage error	584 (5.6)
Death	1221 (4.2)	Death	514 (4.9)
Hospitalization	733 (2.5)	Off label use	428 (4.1)
Dyspnea	661 (2.3)	Circumstance or information capable of leading to medication error	232 (2.2)
Nausea	621 (2.1)	Hemoglobin decreased	206 (2.0)
Vomiting	532 (1.8)	Drug ineffective	163 (1.6)
Pruritus	478 (1.6)	Anemia	150 (1.4)
Off label use	415 (1.4)	Hospitalization	139 (1.3)
Product storage error	386 (1.3)	Diarrhea	113 (1.1)
Hemoglobin decreased	311 (1.1)	Asthenia	104 (1.0)

Only preferred terms with a frequency > 1% are presented.

identified 10 robust safety signals significantly associated with long-acting ESAs: back pain, chest pain, drug hypersensitivity, dyspnea, falls, hypotension, pruritus, UTI, urticaria, and vomiting. Importantly, these 10 signals demonstrated high reliability, remaining completely consistent across all sensitivity analyses that reclassified uncertain ESA types (Table 5, Supplementary Table 2, Supplemental Digital Content 1, http://links.lww.com/JPS/A821). Broadening the criteria to include ESAs as either PS or SS expanded the number of identified safety signals to 23. These 23 signals also demonstrated consistency across the sensitivity analyses (Supplementary Table 2, Supplemental Digital Content 1, http://links.lww.com/JPS/A821). While this broader data set captured more systemic and severe events—such as multiple organ dysfunction syndrome, sepsis, and ventricular fibrillation—it simultaneously diluted specific clinical risks. Crucially, falls and hypotension were detected exclusively in the PS-restricted primary analysis and failed to reach the signal threshold in the broader secondary analysis. Lastly, to ensure methodological rigor, this study fully adhered to the READUS-PV checklist (Supplementary Table 3, Supplemental Digital Content 1, http://links.lww.com/JPS/A821).

DISCUSSION

Our disproportionality analysis aimed to evaluate the safety profiles of long-acting versus short-acting ESAs by contrasting specific drug-role analyses (PS versus SS) to detect both known and unlisted safety signals. Regarding CV-related AEs, ventricular fibrillation was the only strict safety signal identified, and it was detected only in the secondary analysis. Among the AEs flagged in the primary analysis, chest pain was identified; however, given that chest pain can occur from various noncardiac causes and may have limited clinical significance even when associated with cardiac issues, it does not definitively point to a strictly CV signal. A review of the U.S. FDA-approved package inserts for the 5 commercially available ESAs indicates that CV risks may increase when targeting hemoglobin levels above 11 g/dL, as explicitly stated in the black box warning.[30–34] Key CV events of interest included death, myocardial infarction, stroke, venous thromboembolism, thrombosis of vascular access. While prior studies have consistently highlighted concerns regarding increased mortality and CV risks associated with long-acting ESAs, our findings suggest that the evidence for a CV safety signal attributable to long-acting ESAs in this database remains weak.[14,15]

Our analysis successfully detected known adverse events —specifically UTI and back pain—associated with long-acting ESAs, which is consistent with current FDA-approved package inserts. Both UTI and back pain are documented as adverse reactions with an incidence rate of ~5% to 6% for methoxy polyethylene glycol-epoetin beta.[30–34] The successful identification of these established safety signals reinforces the reliability of our disproportionality analysis in capturing true drug-event associations from the FAERS database.

While UTI is a documented risk, managing it in the primary ESA user population is clinically complex. ESAs are

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TABLE 4. Safety Signals of Long-Acting Compared With Short-Acting ESAs

	No. Reports		Disproportionality Analysis			
System Organ Class and Preferred Terms	**Long**	**Short**	**PRR**	**ROR**	**IC025**	**χ^2**
Primary analysis (ESA as primary suspect drug)						
Back pain	84	6	5.8 (2.5-13.2)	5.8 (2.5-13.3)	0.04	21.2
Chest pain	108	9	5.0 (2.5-9.8)	5.0 (2.5-9.9)	0.06	25.4
Drug hypersensitivity	551	7	32.5 (15.4-68.5)	34.1 (16.1-71.8)	0.34	217.1
Dyspnea	323	37	3.6 (2.6-5.1)	3.7 (2.6-5.2)	0.16	63.0
Fall	151	19	3.3 (2.0-5.3)	3.3 (2.1-5.3)	0.06	26.2
Hypotension	109	10	4.5 (2.4-8.6)	4.5 (2.4-8.7)	0.05	24.1
Pruritus	223	21	4.4 (2.8-6.9)	4.5 (2.8-7.0)	0.15	49.9
Urinary tract infection	162	15	4.5 (2.6-7.6)	4.5 (2.7-7.7)	0.11	36.3
Urticaria	73	3	10.1 (3.2-31.9)	10.1 (3.2-32.1)	0.05	22.4
Vomiting	132	18	3.0 (1.9-5.0)	3.1 (1.9-5.0)	0.03	20.9
Secondary analysis (ESA as primary or secondary suspect drug)						
Abdominal distension	156	5	11.1 (4.6-27.1)	11.2 (4.6-27.3)	0.13	43.7
Ascites	146	4	13.0 (4.8-35.2)	13.1 (4.8-35.4)	0.13	42.2
Back pain*	175	13	4.8 (2.7-8.4)	4.8 (2.7-8.5)	0.09	35.7
Chest pain*	171	19	3.2 (2.0-5.2)	3.2 (2.0-5.2)	0.03	25.4
Constipation	182	14	4.6 (2.7-8.0)	4.7 (2.7-8.0)	0.09	36.4
Drug hypersensitivity*	1,740	19	32.7 (20.8-51.4)	34.7 (22.1-54.6)	0.35	603.1
Dyspnea*	661	74	3.2 (2.5-4.1)	3.2 (2.5-4.1)	0.16	101.1
Erythema	105	5	7.5 (3.1-18.4)	7.5 (3.1-18.5)	0.05	25.9
Feeling hot	155	7	7.9 (3.7-16.9)	7.9 (3.7-17.0)	0.11	39.5
Flushing	263	3	31.3 (10.0-97.7)	31.6 (10.1-98.6)	0.22	86.3
General physical health deterioration	166	13	4.6 (2.6-8.0)	4.6 (2.6-8.1)	0.07	32.7
Hyponatremia	145	10	5.2 (2.7-9.8)	5.2 (2.7-9.9)	0.07	30.6
Inappropriate schedule of product administration	198	8	8.8 (4.4-17.9)	8.9 (4.4-18.0)	0.15	52.6
Multiple organ dysfunction syndrome	154	4	13.8 (5.1-37.1)	13.8 (5.1-37.3)	0.14	45.1
Nausea	621	63	3.5 (2.7-4.6)	3.6 (2.8-4.6)	0.17	104.1
No adverse event	221	7	11.3 (5.3-23.9)	11.4 (5.3-24.1)	0.17	62.7
Pruritus*	478	70	2.4 (1.9-3.1)	2.5 (1.9-3.2)	0.09	51.9
Sepsis	254	28	3.2 (2.2-4.8)	3.3 (2.2-4.8)	0.08	38.5
Unresponsive to stimuli	147	3	17.5 (5.6-54.9)	17.6 (5.6-55.2)	0.14	44.7
Urinary tract infection*	210	29	2.6 (1.8-3.8)	2.6 (1.8-3.8)	0.03	24.2
Urticaria*	140	9	5.6 (2.8-10.9)	5.6 (2.8-10.9)	0.07	30.7
Ventricular fibrillation	99	3	11.8 (3.7-37.2)	11.8 (3.7-37.3)	0.06	27.6
Vomiting*	532	78	2.4 (1.9-3.1)	2.5 (1.9-3.1)	0.10	57.8

*Safety signals that were identified in both primary and secondary analysis.
IC025 indicates lower limit of the 95% 2-sided CI of the information component; N, number; χ^2, Chi-squared.

predominantly prescribed for anemia in patients with CKD or cancer patients undergoing chemotherapy—populations that often have significantly compromised renal function and an inherently high susceptibility to infections. Consequently, treating UTIs in these patients requires careful antimicrobial dose adjustments based on kidney function. Improper management can lead to uncontrolled infections or further renal deterioration, both of which are associated with poor clinical outcomes such as prolonged hospital stays and increased mortality.[35–38] In contrast to UTIs, which present evident infectious challenges, the etiology and clinical implications of back pain during ESA therapy remain poorly understood. Further research is needed to elucidate the potential mechanisms by which long-acting ESAs induce back pain and to determine its broader clinical relevance for patients undergoing therapy.

Our study identified falls as a novel safety signal, as they are not currently listed as an AE in any ESA drug labels.[30–34] We hypothesize 2 potential mechanisms for this association: falls may be a consequence of insufficient anemia management or a secondary effect of drug-induced hypotension. First, falls may result from insufficient effectiveness in raising hemoglobin, as evidence suggests that patients with low hemoglobin levels are more prone to falling.[39] For long-acting ESAs specifically, this reduced efficacy could be linked to their PEGylated formulations; PEGylation can induce anti-PEG antibodies,[11] which may lead to a inactivation of the drug and subsequent failure to adequately manage anemia. Second, falls may occur as a direct consequence of hypotension, which is a well-documented AE of all long-acting ESAs. Hypotension causes symptoms such as dizziness, lightheadedness, and impaired balance, all of which increase the risk of falls. Both of these hypothesized pathways are particularly critical for vulnerable populations, such as hemodialysis patients, who are already at an increased risk for falling.[40] Further research is warranted to explore whether adequate anemia correction can reduce the risk of falls and hypotension, or if long-acting ESAs inherently pose a higher risk compared with short-acting ESAs.

Dyspnea, pruritus, urticaria, and vomiting, together with drug hypersensitivity, also showed safety signals for long-acting ESAs in our analysis. Theoretically, long-acting ESAs do not inherently cause more drug hypersensitivity reactions than short-acting ESAs based on their active ingredient alone. Evidence indicates that PEGylated drugs can induce the production of antipolyethylene glycol immunoglobulins (IgG and IgE), which may trigger IgE-

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TABLE 5. Comparison of Safety Signals of Long-Acting ESAs Versus Short-Acting ESAs Across Different Classification Scenarios

	Primary Analysis			Secondary Analysis		
	Main	**Sensitivity 1***	**Sensitivity 2†**	**Main**	**Sensitivity 1***	**Sensitivity 2†**
Abdominal distension	X	X	X	O	O	O
Arteriovenous fistula site complication	X	X	X	X	X	O
Ascites	X	X	X	O	O	O
Back pain	O	O	O	O	O	O
Cardiac arrest	X	X	X	X	X	O
Chest discomfort	X	X	X	X	X	O
Chest pain	O	O	O	O	O	O
Constipation	X	X	X	O	O	O
Drug hypersensitivity	O	O	O	O	O	O
Dyspnea	O	O	O	O	O	O
Erythema	X	X	X	O	O	O
Fall	O	O	O	X	X	X
Feeling hot	X	X	X	O	O	O
Flushing	X	X	X	O	O	O
General physical health deterioration	X	X	X	O	O	O
Hemodialysis	X	X	X	X	X	O
Hospitalization	X	X	X	X	X	O
Hyponatremia	X	X	X	O	O	O
Hypotension	O	O	O	X	X	X
Inappropriate schedule of product administration	X	X	X	O	O	O
Multiple organ dysfunction syndrome	X	X	X	O	O	O
Nausea	X	X	X	O	O	O
No adverse event	X	X	X	O	O	O
Pruritus	O	O	O	O	O	O
Sepsis	X	X	X	O	O	O
Swelling	X	X	X	X	X	O
Unresponsive to stimuli	X	X	X	O	O	O
Urinary tract infection	O	O	O	O	O	O
Urticaria	O	O	O	O	O	O
Ventricular fibrillation	X	X	X	O	O	O
Vomiting	O	O	O	O	O	O

*An analysis in which uncertain types of ESAs are classified as long-acting ESAs.
†An analysis in which uncertain types of ESAs are classified as short-acting ESAs.

mediated allergic reactions or complement-mediated hypersensitivity (known as complement activation-related pseudoallergy).[11] This biological mechanism aligns closely with recent pharmacovigilance studies, including a previous analysis of the FAERS database, which found that immunogenicity-related adverse events such as pruritus, rash, and erythema, as well as broader hypersensitivity reactions, are more frequently reported for PEGylated medicinal products compared with non-PEGylated ones. Given these findings, a formal comparative evaluation of drug hypersensitivity risks—potentially incorporating screening for anti-PEG antibodies to identify susceptible patients—could provide valuable insights to guide the choice of an appropriate initial ESA treatment and ensure patient safety.

To the best of our knowledge, this study is among the first to evaluate safety signals based on both the duration of ESA action and the specific drug roles using the FAERS database. A major strength of our approach was the use of a PS-specific analysis, which enabled the successful identification of a novel safety signal: falls. Furthermore, our sensitivity analyses effectively confirmed the consistency of these findings by incorporating AE reports involving uncertain ESA types.

However, this study is subject to several limitations inherent to spontaneous AE reporting system, like FAERS. First, there is a high likelihood of underreporting, particularly for less serious AEs. Second, missing data and reporting error may lead to the incomplete capture of ESA-related events. Third, although the reporting system collects highly standardized data using established coding terminologies like MedDRA, these spontaneous submissions lack formal clinical adjudication; therefore, the reported events and diagnoses are not systematically verified by medical experts.[9,15] Consequently, this disproportionality analysis can highlight potential safety signals, but it cannot definitively establish a causal relationship between long-acting ESAs and the detected AEs.

CONCLUSION

This study identified a new safety signal for long-acting ESAs, falls, by evaluating safety according to the duration of action and drug roles of ESAs. This signal was not detected in the broader analysis where drug roles are PS or SS, indicating that accounting for drug roles may be important when analyzing the FAERS database. Furthermore, despite existing concerns and black box warnings regarding cardiovascular risks, our analysis found weak evidence for CV safety signals strictly attributable to long-acting ESAs. Further research, such as cohort studies utilizing large-scale clinical data, is needed to confirm the association between identified signals and long-acting ESAs and to enhance understanding of long-acting ESA safety profiles.

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